

Connor Savugot

B.S. Computer Engineering | North Carolina State University | May 2026

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TECHNICAL PROFILE

Embedded firmware engineer delivering programmable power-conversion control, BMS and IC integration, Embedded Linux services, board bring-up, and protocol-level debugging from architecture and datasheet evaluation through custom-PCB validation.

TECHNICAL SKILLS

Firmware & Systems: C, C++, Python, Bash; FreeRTOS, Embedded Linux, Yocto/Arago, systemd

Hardware & Interfaces: TI TMS320, TI AM62, MSP430FR2355; I2C, SPI, UART/RS232, CAN/CAN FD, ADC, PWM, GPIO, IPMI/IPMB

Development & Test: Git/GitLab, Code Composer Studio, KiCad, SocketCAN/PCAN, JTAG, oscilloscopes, logic/protocol analyzers, through-hole/SMT soldering

ENGINEERING EXPERIENCE

AEGIS Power Systems

Embedded Firmware Engineer

Present

- Developed C firmware for custom bidirectional buck-boost power supplies across transformer-coupled stages using 12 PWM generator pairs (24 physical A/B outputs); independently configured each A output's switching period/frequency, duty cycle, and relative phase, then set its paired B output to match or exactly invert A.
- Evaluated BMS and power-management ICs against electrical, protocol, packaging, and sourcing constraints, then carried selected devices through breakout-board testing, driver/peripheral bring-up, custom-PCB integration, and validation.
- Implemented and debugged CAN/CAN FD, I2C, SPI, and UART interfaces; isolated firmware, timing, wiring, configuration, and physical-layer faults with register-level tests, scopes, analyzers, known-good nodes, and controlled substitutions.
- Supported TI AM62 Yocto/Arago BSPs and device trees; wrote systemd and external-watchdog services, built Python/Bash diagnostics, and validated IPMI/IPMB sensor and FRU access.

Engineering Intern

May–Aug 2024 & May–Aug 2025

- Built a restricted SSH/chroot command interface and Python/Bash tools for approved remote control and inspection of TI Yocto/Arago embedded power hardware.
- Developed packetized, checksum-validated TMS320 flashing utilities, reverse-engineered RS232 command/checksum transactions from limited documentation, and supported FreeRTOS, serial-interface, schematic, and burn-in test debugging.

TEAM Industries | Engineering Intern

Jun–Aug 2021 & Jun–Aug 2022

- Built an Excel/VBA tolerance checker for out-of-spec manufactured parts and organized a searchable 17,000+ record catalog linking tooling, parts, vendors, operations, locations, and revision data.

SYSTEMS DESIGN

EV Active Sensor Adapter | MSP430FR2355

NC State University

- Defined architecture, requirements, interface specifications, and validation plans for an embedded system spanning sensing, BMS/power-management hardware, custom-PCB interfaces, communications, and telemetry.
- Evaluated BMS/support ICs and prototyped interrupt-driven SPI, ADC, UART, timer, GPIO, and LCD interfaces while coordinating integration through trade studies, risk tracking, design reviews, and subsystem test plans.

LEADERSHIP & TEACHING

Embedded Systems Teaching Assistant | NC State University

- Delivered supplemental lectures and laboratory support for C/MSP430 firmware, ADC, timers, interrupts, GPIO, LCDs, and custom-PCB bring-up; taught systematic diagnosis of code, wiring, soldering, and toolchain faults.

Lead Instructor | IEEE Open Project Space 2

Two Semesters

- Led a ten-week, project-based electronics course from scope through final demonstration; taught through-hole/SMT soldering with custom IEEE PCBs and supported student development using PlatformIO, GitHub, KiCad, and BOM workflows.